



TDEP : Target-perturbed Differential Expression Profiles for Interpretable Drug-Target Interaction Prediction

Jieun Sung¹, Sookyung Kim², and Wankyu Kim^{1*}

¹ Department of Life Science, Ewha Womans University

² Department of Artificial Intelligence, Ewha Womans University

* For whom the correspondence should be: wkimwkim@gmail.com



Abstract

Understanding drug-target interactions (DTIs) is a crucial step in drug discovery. While recent AI approaches have accelerated this process, most existing models remain limited to binary predictions -whether a drug interacts with a target- without explaining how such interactions affect cellular states. Here, we present **TDEP (Target-perturbed Differential Expression Profile)**, an AI framework that not only predicts DTIs with accuracy comparable to state-of-the-art methods but also provides interpretable profiles of target protein perturbation effects. These profiles reveal shared patterns among interacting proteins, highlight pathway-level distinctions across target groups, and show strong concordance with genetic perturbation experiments. TDEP also generalizes across unseen cell types by leveraging basal expression similarity, maintaining robust predictive performance. Beyond prediction, TDEP enables comparisons with disease transcriptomes to infer potential therapeutic targets. This interpretability and translatability position TDEP as a practical tool for both mechanistic understanding and real-world drug discovery applications.

Materials & Methods

Comparison of Drug-Target Interaction Prediction Models

Category	Model	Input Representation		Predictive Objective	MoA Explainability
		Ligand	Protein		
Structure-based	MolTrans	Graph	Sequence	DTI only	Low
	GraphDTA	SMILES	Sequence		
	DrugBAN	Graph	Sequence		
Omics-based	ssGCN	CMAP	CMAP	DTI only	Moderate
	FRoGS	CMAP	CMAP		
	TDEP	CMAP	Sequence	DTI, Target-perturbed expression	High

Table 1. Classification and characterization of TDEP and other AI-based DTI models

Overall framework of TDEP (Target-perturbed Differential Expression Profile)

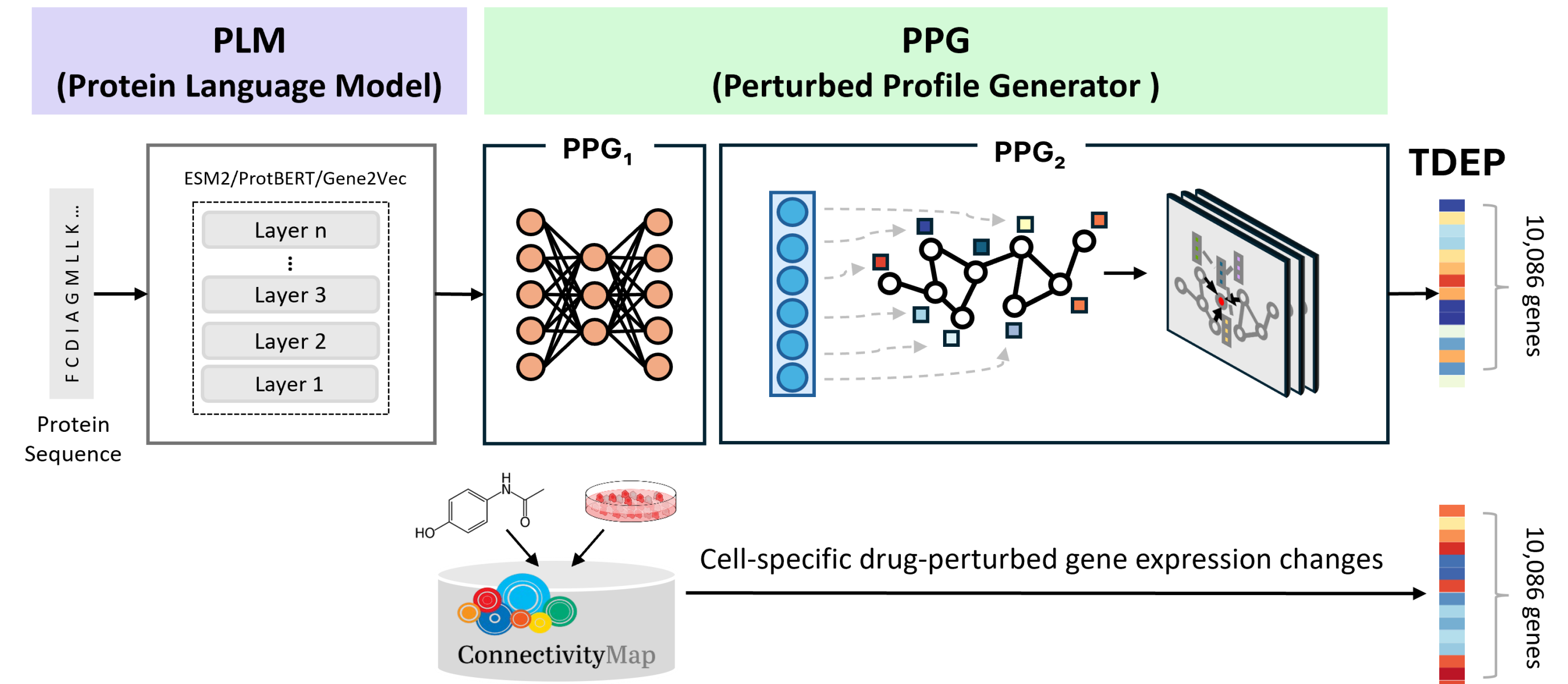


Figure 1. TDEP : Target-perturbed Differential Expression Profile

Results

I. Comparative Evaluation of TDEP with existing models for DTI Prediction

Model	AUROC	AUPRC	F1	Precision@20%Recall	Precision@50%Recall
TDEP _{ESM2}	0.846 ± 0.064	0.827 ± 0.092	0.794 ± 0.051	0.550 ± 0.012	0.644 ± 0.015
TDEP _{ProtBERT}	0.833 ± 0.063	0.816 ± 0.090	0.787 ± 0.052	0.549 ± 0.012	0.640 ± 0.014
TDEP _{Gene2Vec}	0.773 ± 0.124	0.769 ± 0.133	0.729 ± 0.111	0.535 ± 0.026	0.617 ± 0.040
ssGCN	0.595 ± 0.131	0.610 ± 0.122	0.579 ± 0.113	0.516 ± 0.032	0.564 ± 0.070
FRoGS	0.525 ± 0.015	0.521 ± 0.014	0.520 ± 0.010	0.507 ± 0.004	0.518 ± 0.010
DrugBAN	0.836 ± 0.056	0.837 ± 0.067	0.779 ± 0.048	0.543 ± 0.017	0.648 ± 0.035
MolTrans	0.823 ± 0.053	0.856 ± 0.044	0.749 ± 0.049	0.537 ± 0.008	0.607 ± 0.024
GraphDTA _{GATNet}	0.828 ± 0.034	0.860 ± 0.023	0.766 ± 0.028	0.544 ± 0.013	0.620 ± 0.023
GraphDTA _{GATGCN}	0.798 ± 0.049	0.845 ± 0.028	0.729 ± 0.052	0.534 ± 0.018	0.601 ± 0.026
GraphDTA _{GINConvNet}	0.632 ± 0.073	0.636 ± 0.076	0.602 ± 0.063	0.510 ± 0.029	0.565 ± 0.051
GraphDTA _{GCNNet}	0.802 ± 0.037	0.835 ± 0.026	0.741 ± 0.030	0.525 ± 0.016	0.597 ± 0.025

Table 2. Performance of DTIPrediction Models. Values are mean ± standard deviation from six cell lines with balanced positive/negative sampling across 10 random seeds following ssGCN and FRoGS protocols. The best value in each column is highlighted in bold, while the top 3 values are shaded in gray.

II. TDEP Shows Robust Cross-Cell Transferability

	Drug-perturbed expression	TDEP
Same-cell (SC)	MCF7	MCF7
Cross-cell (CC)	MCF7	HA1E
Multi-cell (CC)	MCF7	Weighted sum of five cells excluding MCF7

Table 3. Example input cells for Same-cell (SC), Cross-cell (CC), and Multi-cell (MC) settings in DTI prediction

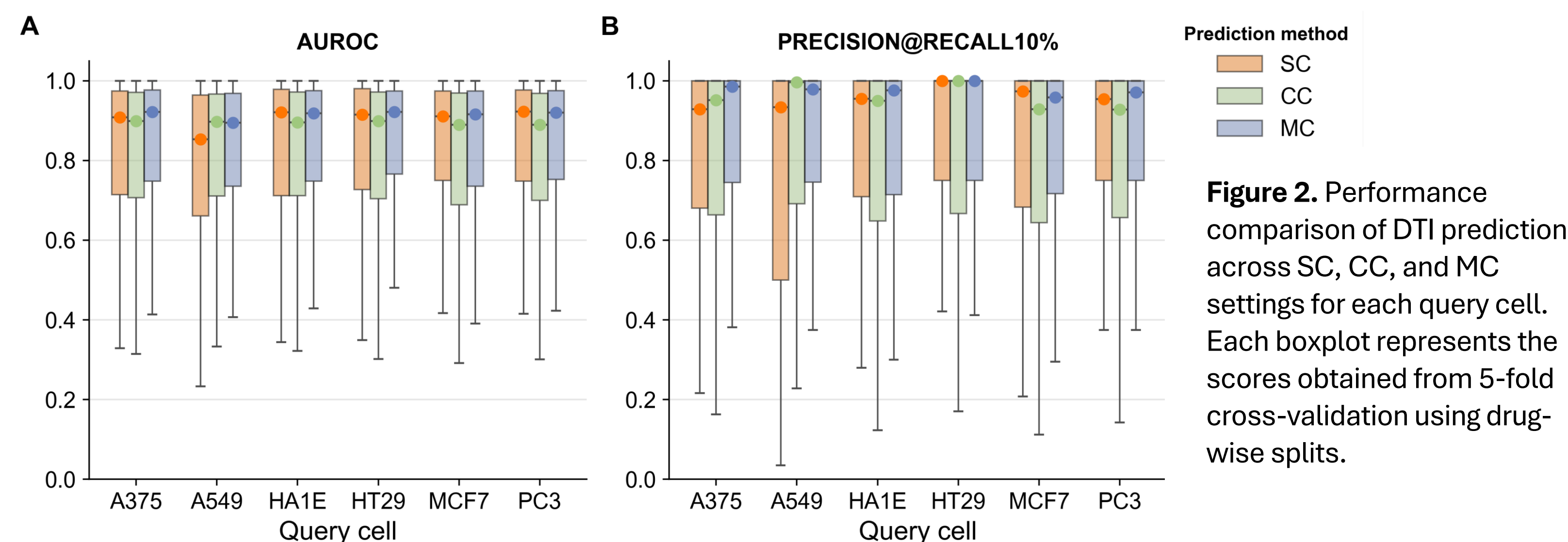


Figure 2. Performance comparison of DTI prediction across SC, CC, and MC settings for each query cell. Each boxplot represents the scores obtained from 5-fold cross-validation using drug-wise splits.

III. TDEP Enables to mimic polypharmacological effect of drugs

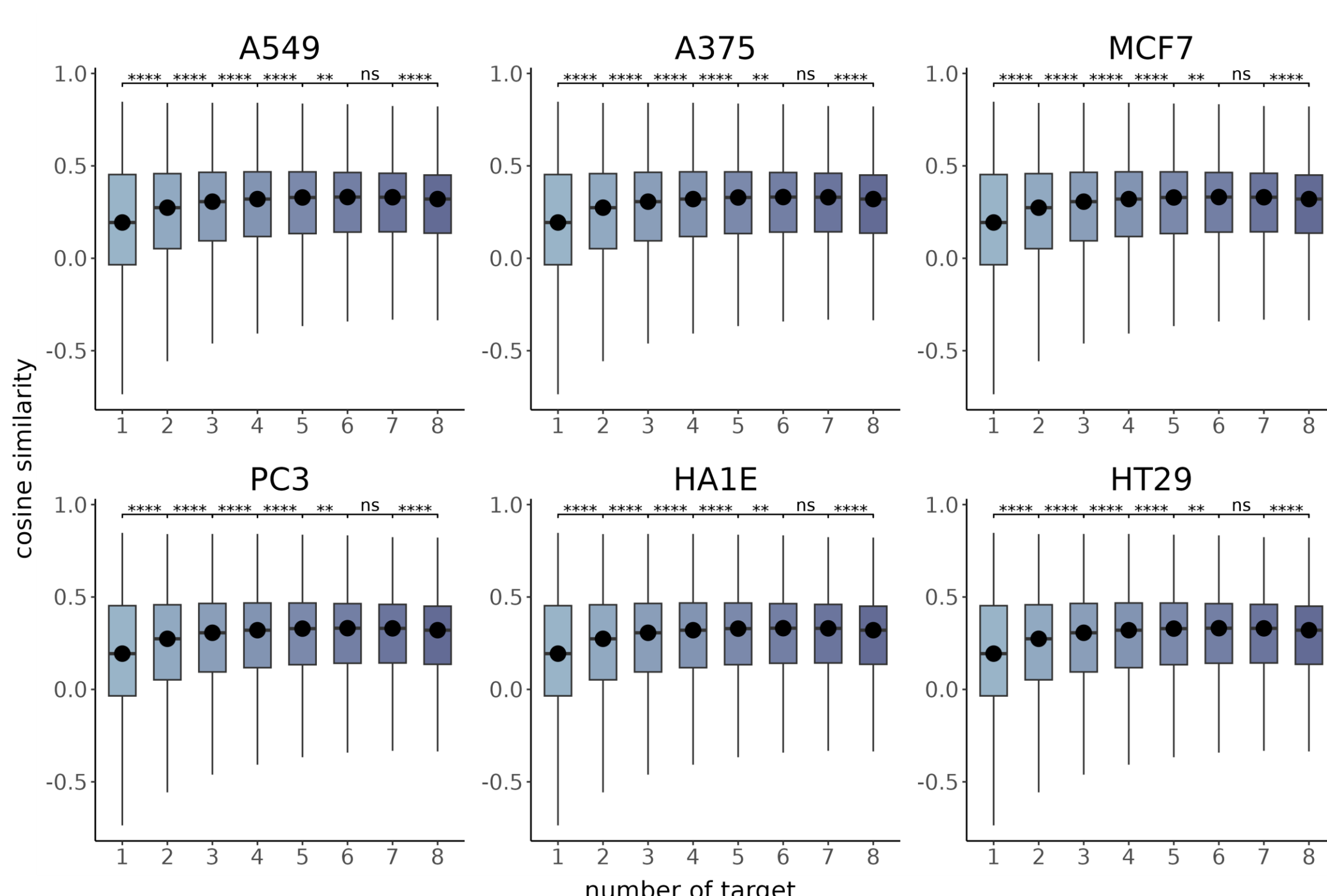


Figure 3. Evaluation of multi-target perturbation effects across six cell lines. It shows cosine similarity between mean-aggregated TDEP profiles and drug-induced signatures. The x-axis shows the number of combined targets; the y-axis shows the cosine similarity between target-aggregated TDEP and drug signatures. (* p ≤ 0.05, ** p ≤ 0.01, *** p ≤ 0.001, **** p ≤ 0.0001; ns: not significant)

IV. TDEP corresponds to shRNA-perturbed signatures

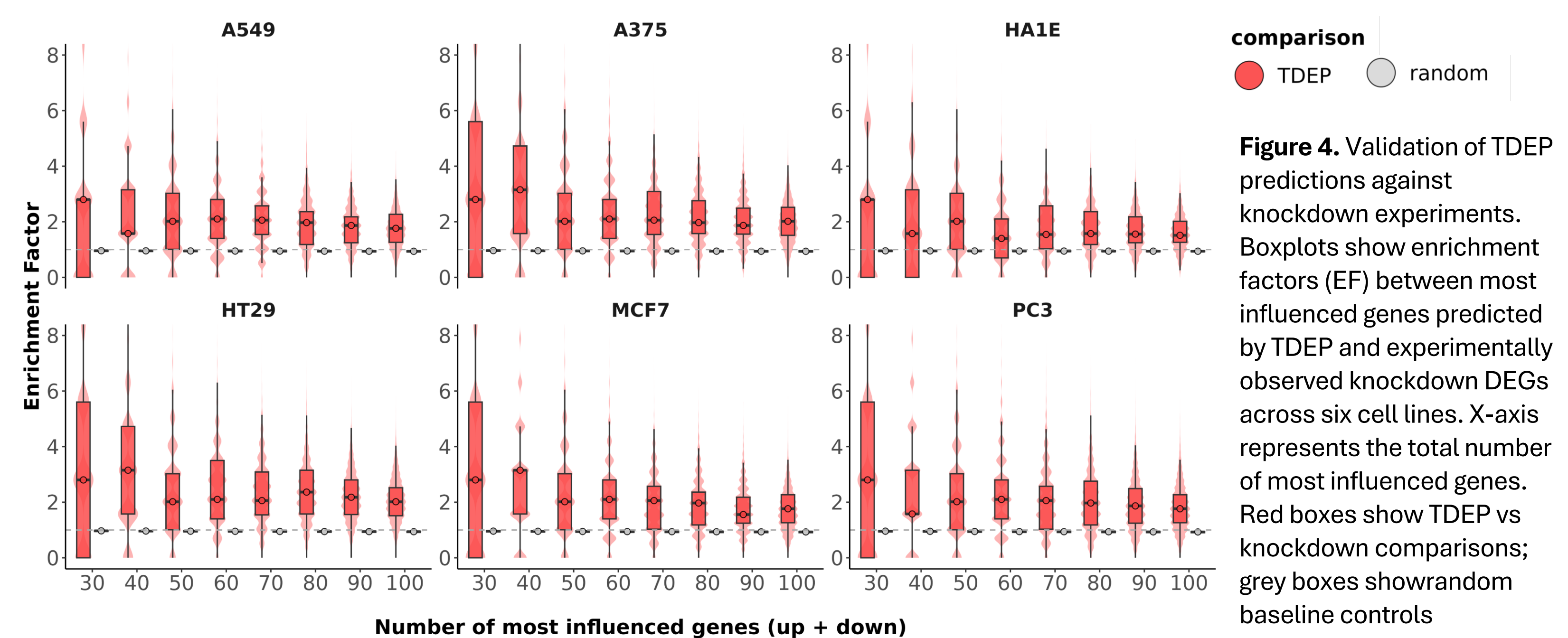


Figure 4. Validation of TDEP predictions against knockdown experiments. Boxplots show enrichment factors (EF) between most influenced genes predicted by TDEP and experimentally observed knockdown DEGs across six cell lines. X-axis represents the total number of most influenced genes. Red boxes show TDEP vs knockdown comparisons; grey boxes show random baseline controls

V. TDEP Predictions Highly Correlate with PROTAC-induced Expression Changes

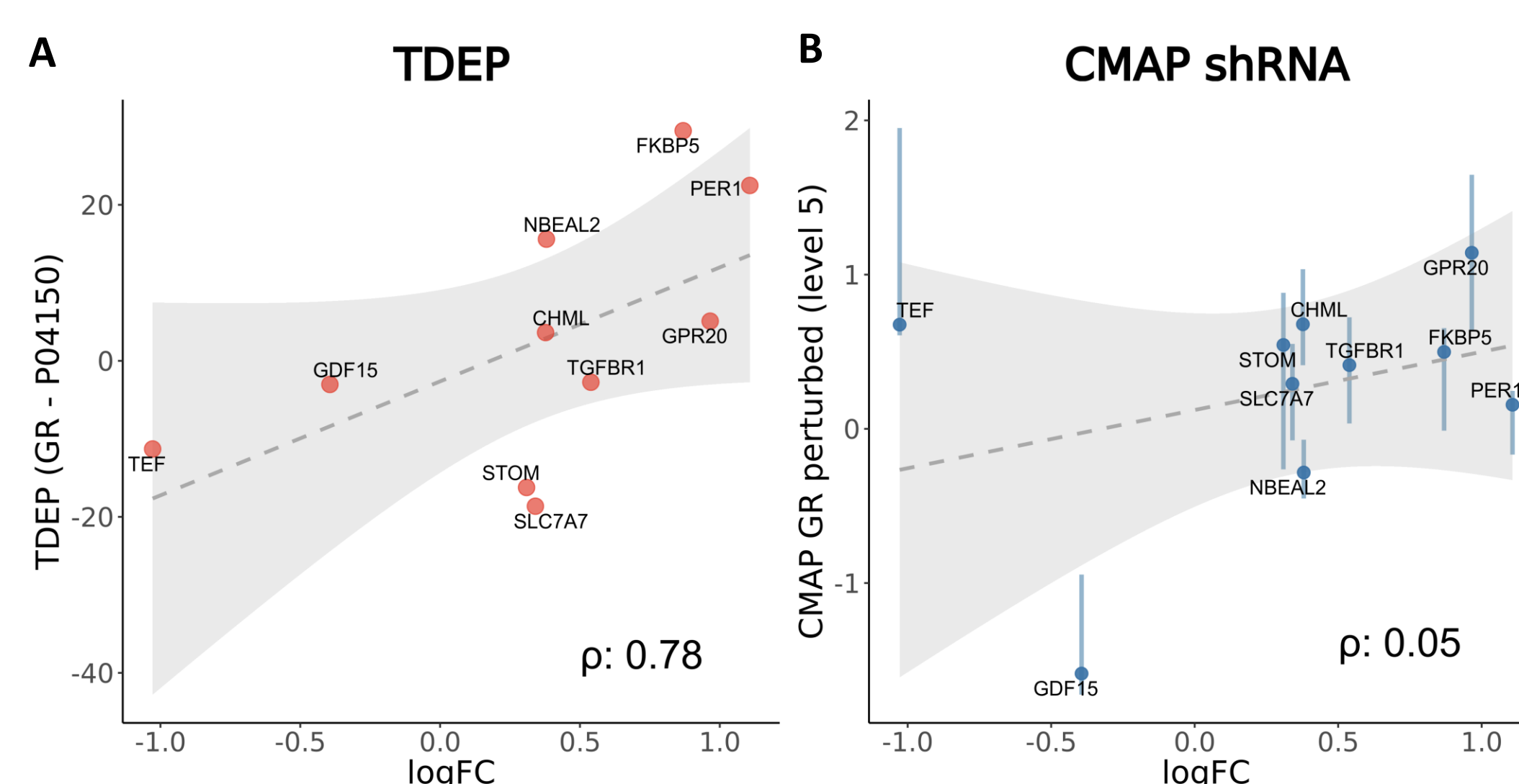


Figure 5. Comparison of TDEP-predicted and experimental PROTAC-induced gene expression changes. PROTAC-treated RNA-seq data were obtained from GSE229084. Differential expression (logFC) was computed using DESeq2 and compared with TDEP predictions (A) and CMAP shRNA-treated profiles (B). Each point represents the top 10 DEGs from PROTAC-treated samples. TDEP predictions showed a strong positive correlation with experimental results (p = 0.78), whereas shRNA-based perturbations showed weak correlations.

VI. TDEP-Based Protein Modules Show Distinct Pathway Patterns

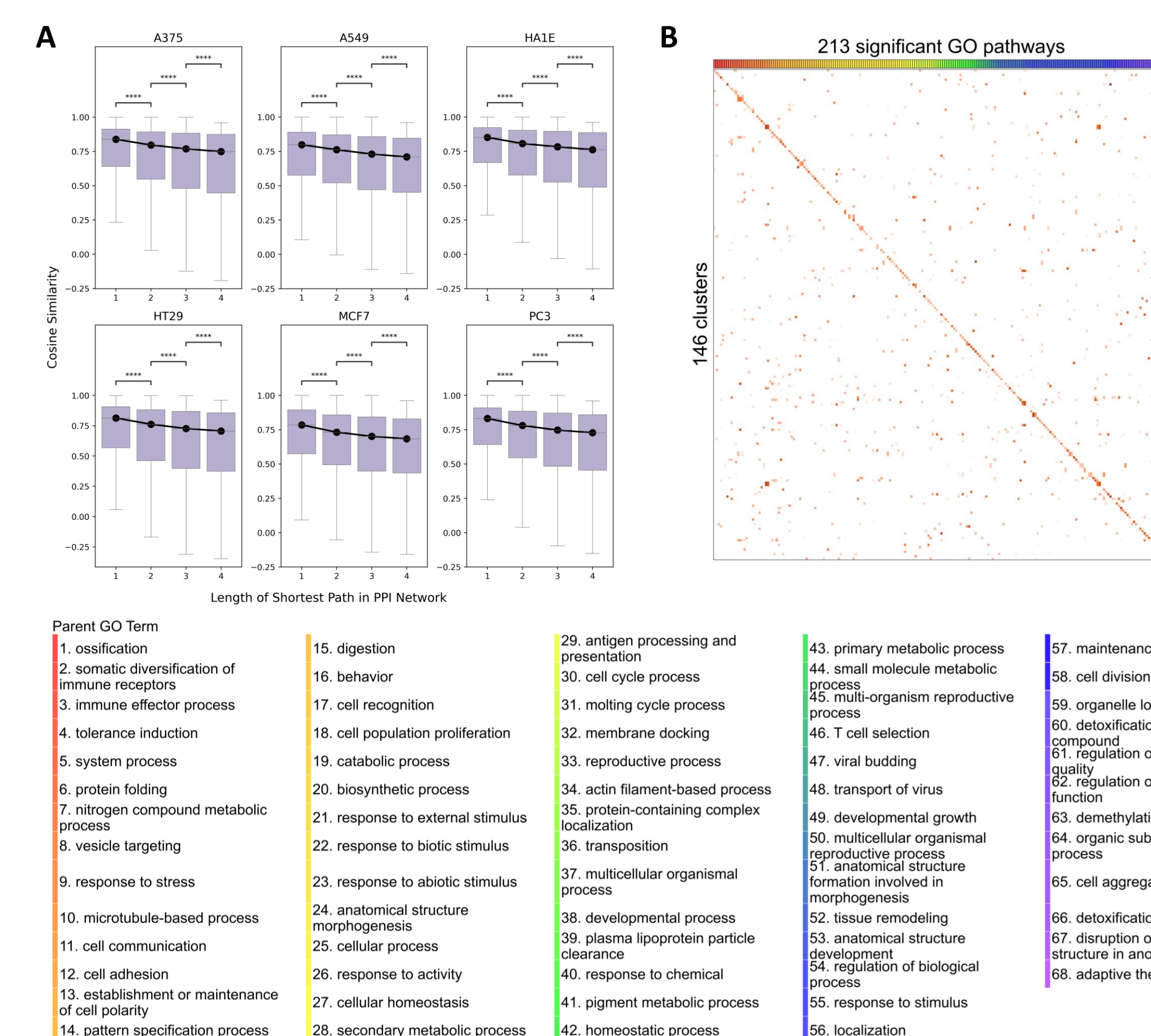


Figure 6A. Relationship between PPI network proximity and TDEP similarity across six cell lines. X-axis: shortest path length in the PPI network; Y-axis: cosine similarity between TDEP profiles. Statistical significance was tested by t-test between adjacent distance groups (* p ≤ 0.05, ** p ≤ 0.01, *** p ≤ 0.001, **** p ≤ 0.0001; ns: not significant)

Figure 6B. Pathway enrichment heatmap of target genes in consensus clusters across six cell lines. Each pathway represents GO level 4, and colors indicate their parent GO level 3 categories.